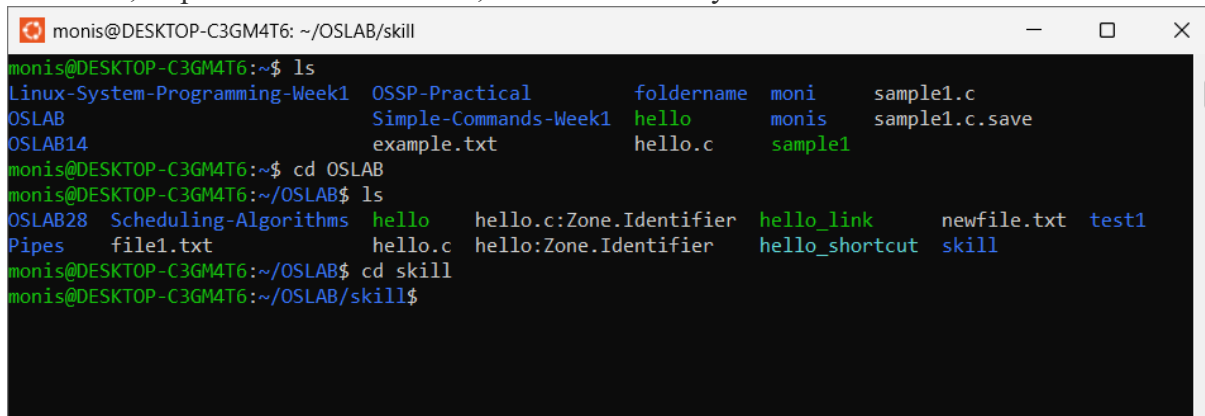


OSSP (Operating Systems And Systems Programming) - 25CS2104E

(SKILL Content Delivery Concepts)

10.To Retrieve Current Directory, Display Path, Process Exit Requests, Cleanup Resources, Save State, Verify Correct Behavior.

To Parse Export Syntax, Update Environment Variables, Validate Names, Handle Existing Variables, Export to Child Processes, Test Functionality.



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~$ ls
Linux-System-Programming-Week1  OSSP-Practical      foldername  moni      sample1.c
OSLAB                          Simple-Commands-Week1 hello       monis     sample1.c.save
OSLAB14                        example.txt         hello.c     sample1
monis@DESKTOP-C3GM4T6:~$ cd OSLAB
monis@DESKTOP-C3GM4T6:~/OSLAB$ ls
OSLAB28  Scheduling-Algorithms  hello    hello.c:Zone.Identifier  hello_link  newfile.txt  test1
Pipes    file1.txt              hello.c  hello:Zone.Identifier    hello_shortcut skill
monis@DESKTOP-C3GM4T6:~/OSLAB$ cd skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 1 — Create the program

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 directory_state.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <limits.h>

int main()
{
    char current_dir[PATH_MAX];

    if (getcwd(current_dir, sizeof(current_dir)) == NULL)
    {
        perror("getcwd");
        return 1;
    }

    printf("Current Directory: %s\n", current_dir);

    printf("Program is running successfully.\n");

    printf("Enter 'exit' to terminate the program.\n");

    char input[50];

    while (1)
    {
        printf("state> ");
        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL)
        {
            break;
        }

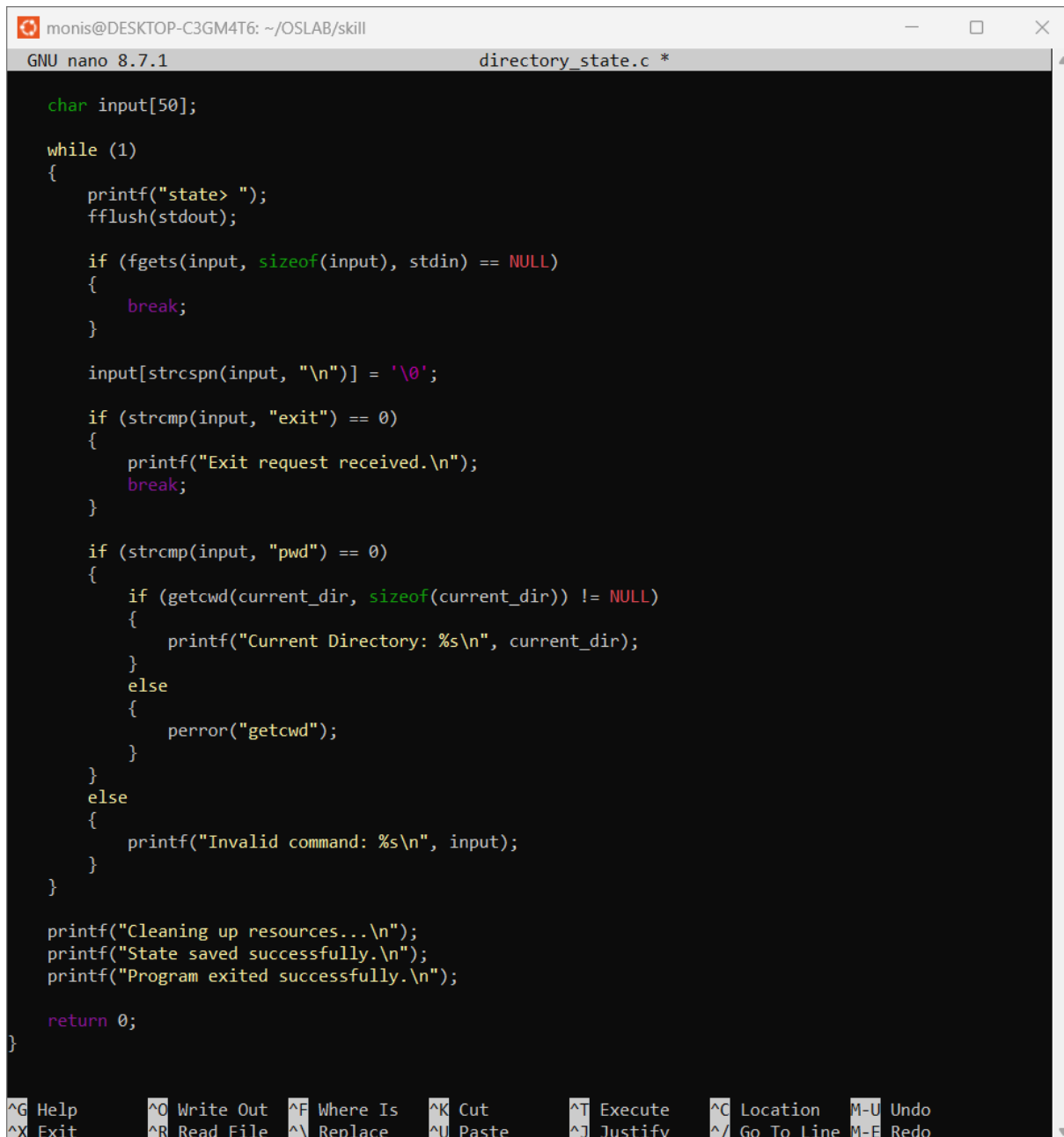
        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0)
        {
            printf("Exit request received.\n");
            break;
        }

        if (strcmp(input, "pwd") == 0)
        {
            if (getcwd(current_dir, sizeof(current_dir)) != NULL)

```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location	M-U Undo
^X Exit	^R Read File	^N Replace	^U Paste	^J Justify	^_ Go To Line	M-E Redo



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 directory_state.c *

char input[50];

while (1)
{
    printf("state> ");
    fflush(stdout);

    if (fgets(input, sizeof(input), stdin) == NULL)
    {
        break;
    }

    input[strcspn(input, "\n")] = '\0';

    if (strcmp(input, "exit") == 0)
    {
        printf("Exit request received.\n");
        break;
    }

    if (strcmp(input, "pwd") == 0)
    {
        if (getcwd(current_dir, sizeof(current_dir)) != NULL)
        {
            printf("Current Directory: %s\n", current_dir);
        }
        else
        {
            perror("getcwd");
        }
    }
    else
    {
        printf("Invalid command: %s\n", input);
    }
}

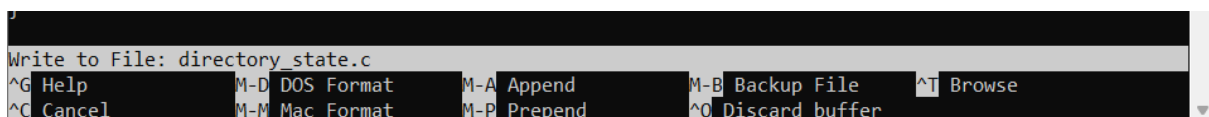
printf("Cleaning up resources...\n");
printf("State saved successfully.\n");
printf("Program exited successfully.\n");

return 0;
}
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo

Save:

Ctrl + O → Enter → Ctrl + X



Write to File: directory_state.c

^G Help M-D DOS Format M-A Append M-B Backup File ^T Browse
^C Cancel M-M Mac Format M-P Prepend ^Q Discard buffer

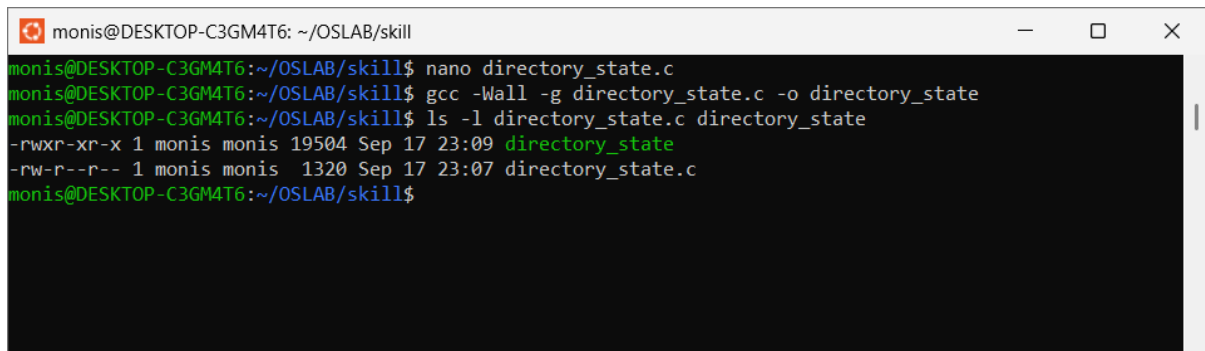
Step 2 — Compile the Program

Run:

`gcc -Wall -g directory_state.c -o directory_state`

Then run:

`ls -l directory_state.c directory_state`

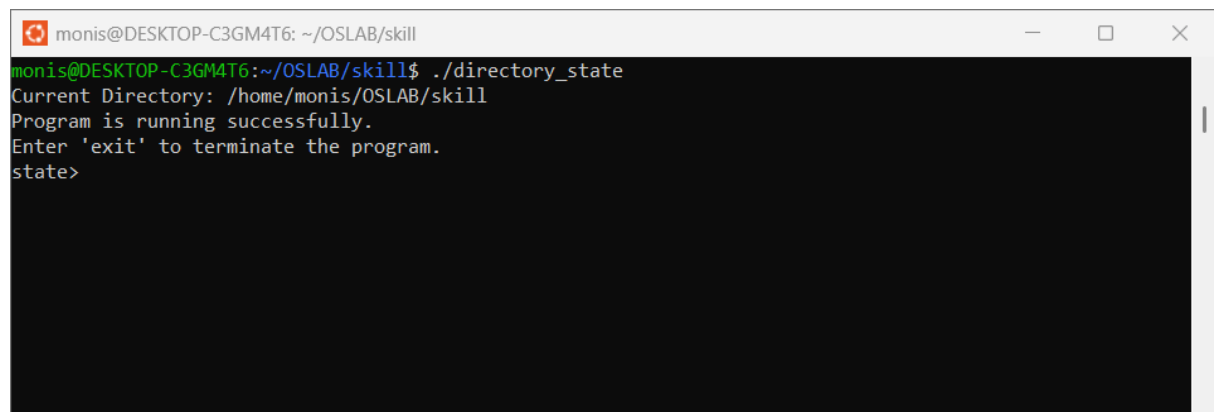


```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano directory_state.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc -Wall -g directory_state.c -o directory_state
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls -l directory_state.c directory_state
-rwxr-xr-x 1 monis monis 19504 Sep 17 23:09 directory_state
-rw-r--r-- 1 monis monis 1320 Sep 17 23:07 directory_state.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 3 — Run the Program and Retrieve the Current Directory

Run:

`./directory_state`



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./directory_state
Current Directory: /home/monis/OSLAB/skill
Program is running successfully.
Enter 'exit' to terminate the program.
state>
```

Step 4 — Test the pwd Function

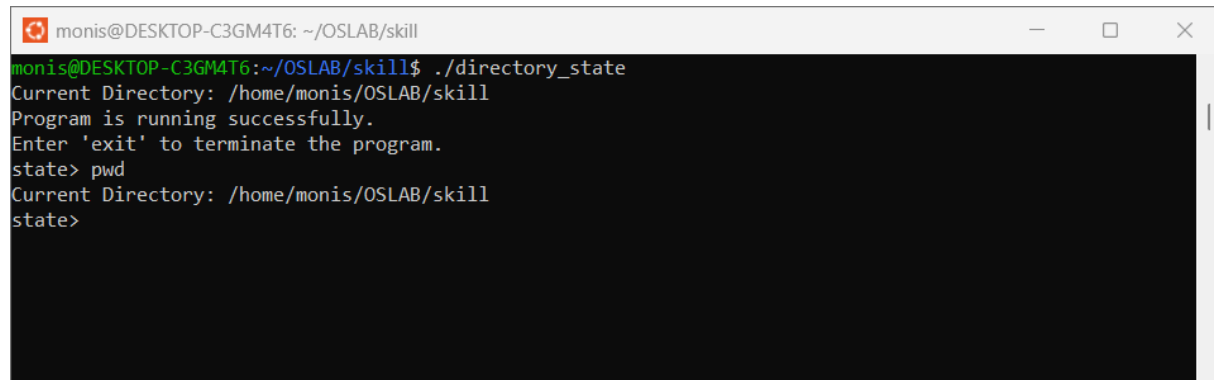
At the prompt:

`state>`

type:

`pwd`

Press **Enter**.

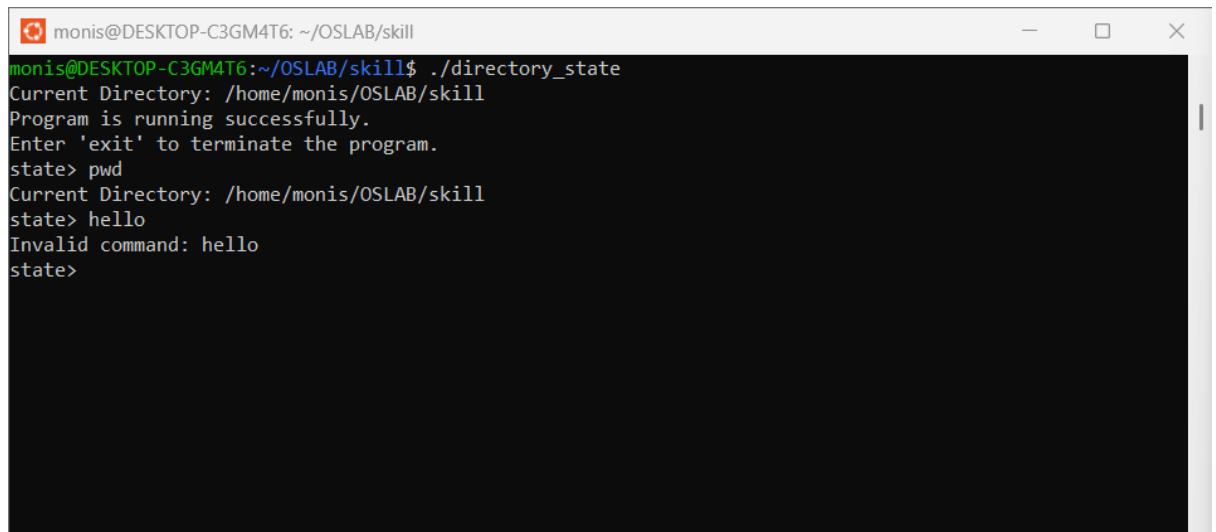


```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./directory_state
Current Directory: /home/monis/OSLAB/skill
Program is running successfully.
Enter 'exit' to terminate the program.
state> pwd
Current Directory: /home/monis/OSLAB/skill
state>
```

Step 5 — Test an Invalid Command

At:

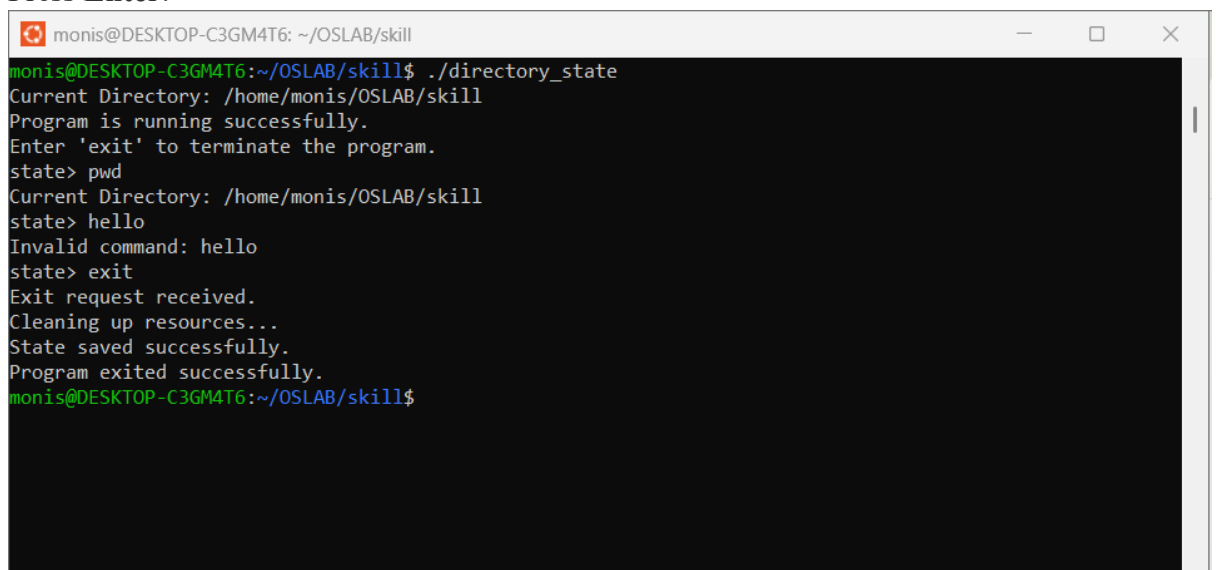
state>
type:
hello
Press **Enter**.



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./directory_state
Current Directory: /home/monis/OSLAB/skill
Program is running successfully.
Enter 'exit' to terminate the program.
state> pwd
Current Directory: /home/monis/OSLAB/skill
state> hello
Invalid command: hello
state>
```

Step 6 — Test the Exit Request

At:
state>
type:
exit
Press **Enter**.



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./directory_state
Current Directory: /home/monis/OSLAB/skill
Program is running successfully.
Enter 'exit' to terminate the program.
state> pwd
Current Directory: /home/monis/OSLAB/skill
state> hello
Invalid command: hello
state> exit
Exit request received.
Cleaning up resources...
State saved successfully.
Program exited successfully.
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 7 — Create the Export Program

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 export_variables.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>
#include <unistd.h>
#include <sys/wait.h>

#define INPUT_SIZE 200

int valid_name(const char *name)
{
    if (name == NULL || name[0] == '\0')
    {
        return 0;
    }

    if (!isalpha((unsigned char)name[0]) && name[0] != '_')
    {
        return 0;
    }

    for (int i = 1; name[i] != '\0'; i++)
    {
        if (!isalnum((unsigned char)name[i]) && name[i] != '_')
        {
            return 0;
        }
    }

    return 1;
}

void process_export(char *input)
{
    char *name;
    char *value;
    char *equals;

    if (strncmp(input, "export ", 7) != 0)
    {
        printf("Invalid command.\n");
        return;
    }

    char *assignment = input + 7;
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location	M-U Undo
^X Exit	^R Read File	^H Replace	^U Paste	^J Justify	^_ Go To Line	M-E Redo

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 export_variables.c *

char *assignment = input + 7;

equals = strchr(assignment, '=');

if (equals == NULL)
{
    printf("Error: Export syntax must be NAME=VALUE.\n");
    return;
}

*equals = '\0';

name = assignment;
value = equals + 1;

if (!valid_name(name))
{
    printf("Error: Invalid variable name: %s\n", name);
    return;
}

if (setenv(name, value, 1) != 0)
{
    perror("setenv");
    return;
}

printf("Environment variable updated.\n");
printf("%s=%s\n", name, value);
}

int main()
{
    char input[INPUT_SIZE];

    printf("Export Environment Shell\n");

    while (1)
    {
        printf("export> ");
        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL)
        {
            ^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
            ^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo
        }
    }
}
```



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 export_variables.c *
}

int main()
{
    char input[INPUT_SIZE];

    printf("Export Environment Shell\n");

    while (1)
    {
        printf("export> ");
        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL)
        {
            break;
        }

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0)
        {
            break;
        }

        if (strcmp(input, "show") == 0)
        {
            char *path = getenv("PATH");

            if (path != NULL)
            {
                printf("PATH=%s\n", path);
            }

            continue;
        }

        process_export(input);
    }

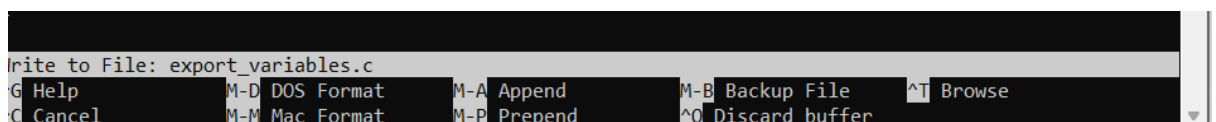
    printf("Export shell exited.\n");

    return 0;
}
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/_ Go To Line M-E Redo

Save the file:

Ctrl + O → Enter → Ctrl + X



Step 8 — Compile the Export Program

Run:

```
gcc -Wall -g export_variables.c -o export_variables
```

If there are **no error messages**, compilation was successful.

Now check the files:

```
ls -l export_variables.c export_variables
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano export_variables.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc -Wall -g export_variables.c -o export_variables
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls -l export_variables.c export_variables
-rwxr-xr-x 1 monis monis 20768 Sep 17 23:17 export_variables
-rw-r--r-- 1 monis monis 1969 Sep 17 23:17 export_variables.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 9 — Run the Export Program

Run:

`./export_variables`

You should see:

Export Environment Shell

export>

Now type:

export NAME=Monisha

Press **Enter**.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./export_variables
Export Environment Shell
export> export NAME=Monisha
Environment variable updated.
NAME=Monisha
export>
```

Step 10 — Test the Updated Environment Variable

The program should still be running at:

export>

Type:

export NAME=Student

Press **Enter**.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./export_variables
Export Environment Shell
export> export NAME=Monisha
Environment variable updated.
NAME=Monisha
export> export NAME=Student
Environment variable updated.
NAME=Student
export>
```

Step 11 — Test Invalid Variable Names

At:

export>

type:

export 123NAME=Test

Press **Enter**.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./export_variables
Export Environment Shell
export> export NAME=Monisha
Environment variable updated.
NAME=Monisha
export> export NAME=Student
Environment variable updated.
NAME=Student
export> export 123NAME=Test
Error: Invalid variable name: 123NAME
export>
```

Step 12 — Test Export Syntax Without =

At:

export>

type:

export TEST

Press **Enter**.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./export_variables
Export Environment Shell
export> export NAME=Monisha
Environment variable updated.
NAME=Monisha
export> export NAME=Student
Environment variable updated.
NAME=Student
export> export 123NAME=Test
Error: Invalid variable name: 123NAME
export> export TEST
Error: Export syntax must be NAME=VALUE.
export>
```

Step 13 — Test Exporting an Empty Value

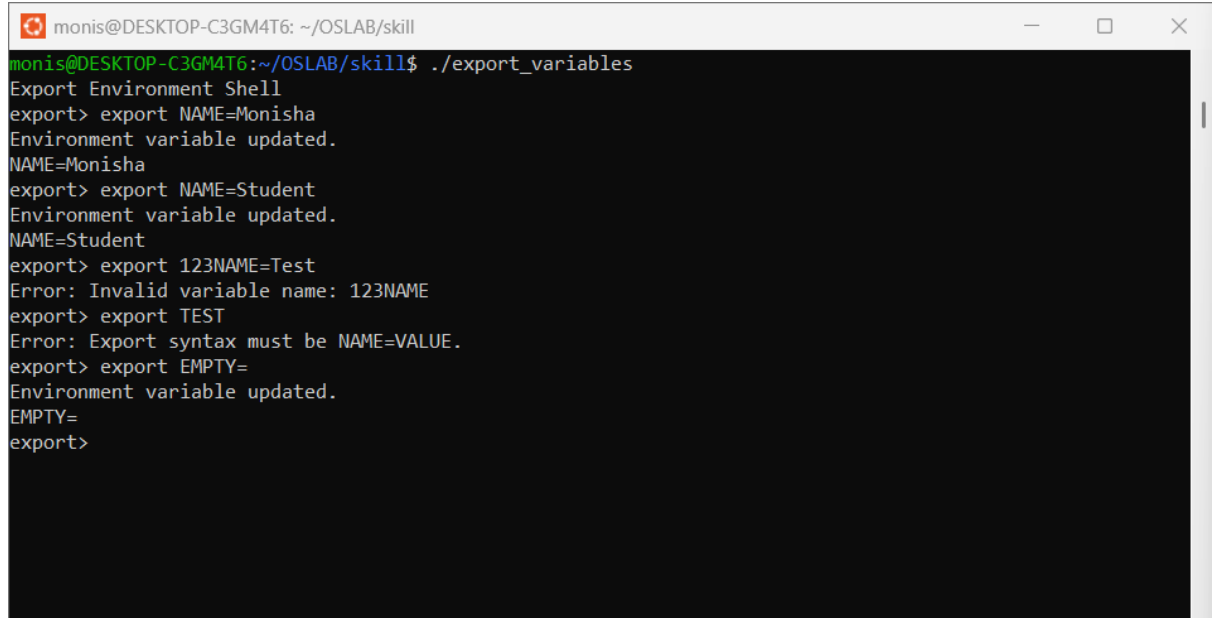
At:

export>

type:

export EMPTY=

Press **Enter**.



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./export_variables
Export Environment Shell
export> export NAME=Monisha
Environment variable updated.
NAME=Monisha
export> export NAME=Student
Environment variable updated.
NAME=Student
export> export 123NAME=Test
Error: Invalid variable name: 123NAME
export> export TEST
Error: Export syntax must be NAME=VALUE.
export> export EMPTY=
Environment variable updated.
EMPTY=
export>
```

Step 14 — Verify the Environment Variable

The program should still be at:

export>

Type:

printenv NAME

But **do not use this inside the running program**—the program does not execute shell commands. Instead, first exit the program:

exit

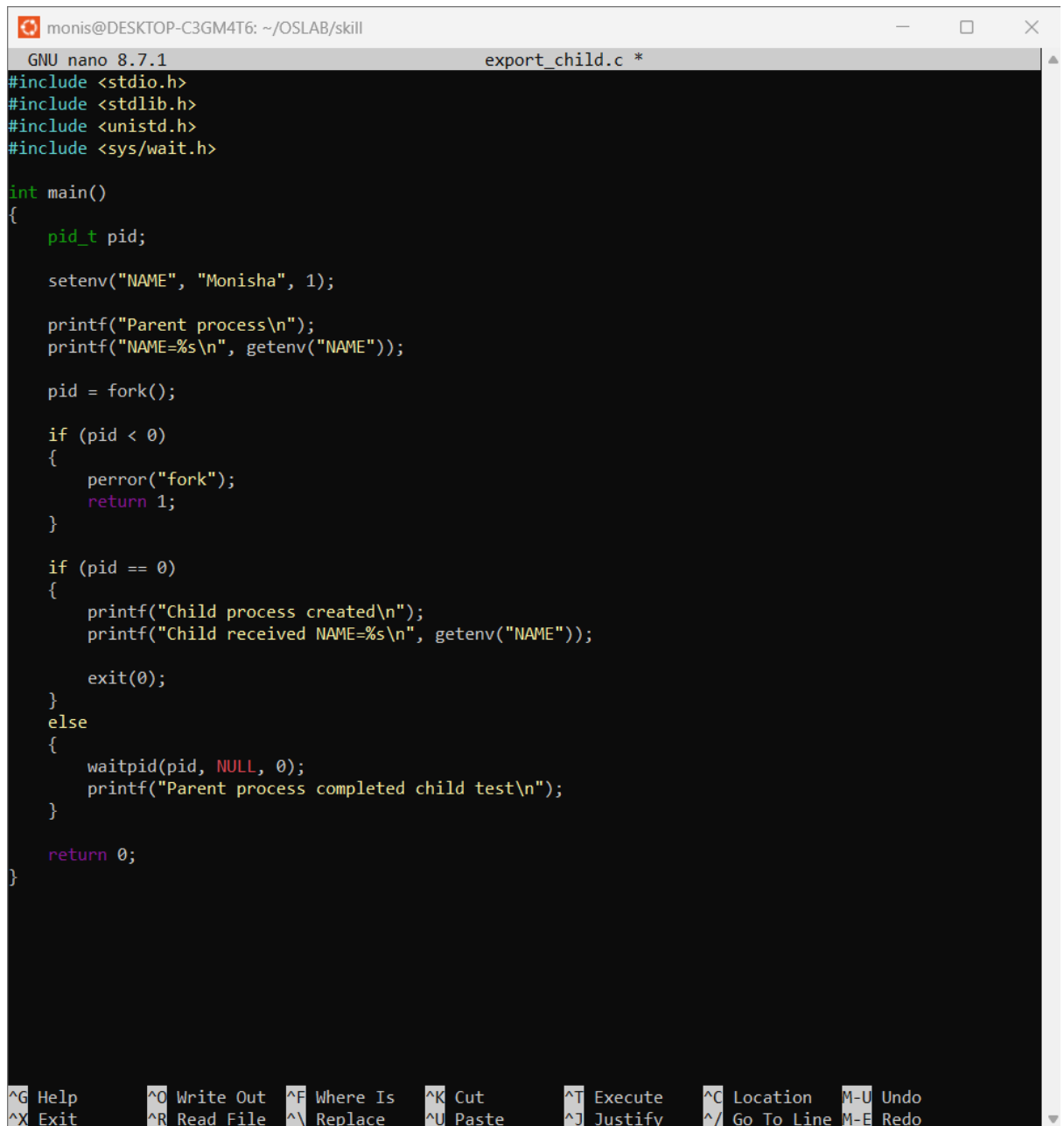
You should see:

Exit shell exited.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill$ ./export_variables
Export Environment Shell
export> export NAME=Monisha
Environment variable updated.
NAME=Monisha
export> export NAME=Student
Environment variable updated.
NAME=Student
export> export 123NAME=Test
Error: Invalid variable name: 123NAME
export> export TEST
Error: Export syntax must be NAME=VALUE.
export> export EMPTY=
Environment variable updated.
EMPTY=
export> ^C
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./export_variables
Export Environment Shell
export> printenv NAME
Invalid command.
export> exit
Export shell exited.
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ printenv NAME
DESKTOP-C3GM4T6
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 15 — Add Child-Process Testing

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill$ nano export_child.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 export_child.c *
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    pid_t pid;

    setenv("NAME", "Monisha", 1);

    printf("Parent process\n");
    printf("NAME=%s\n", getenv("NAME"));

    pid = fork();

    if (pid < 0)
    {
        perror("fork");
        return 1;
    }

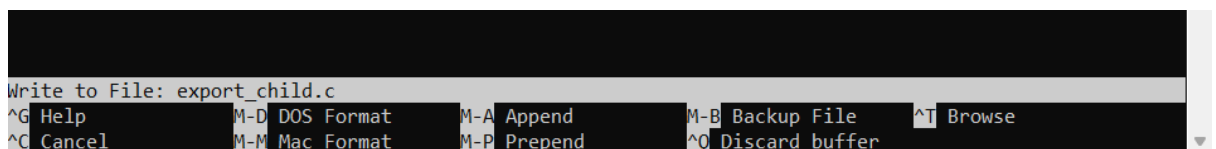
    if (pid == 0)
    {
        printf("Child process created\n");
        printf("Child received NAME=%s\n", getenv("NAME"));

        exit(0);
    }
    else
    {
        waitpid(pid, NULL, 0);
        printf("Parent process completed child test\n");
    }

    return 0;
}
```

Save:

Ctrl + O → Enter → Ctrl + X



Step 16 — Compile the Child-Process Program

Run:

`gcc -Wall -g export_child.c -o export_child`

If there are no error messages, it compiled successfully.

Then check:

`ls -l export_child.c export_child`

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano export_child.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc -Wall -g export_child.c -o export_child
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls -l export_child.c export_child
-rwxr-xr-x 1 monis monis 18040 Sep 17 23:35 export_child
-rw-r--r-- 1 monis monis  588 Sep 17 23:32 export_child.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 17 — Run the Child-Process Test

Now run:

`./export_child`

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./export_child
Parent process
NAME=Monisha
Child process created
Child received NAME=Monisha
Parent process completed child test
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 18 — Test Export Functionality Again

Run:

`./export_variables`

At the prompt, enter:

`export COURSE=OSSP`

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./export_variables
Export Environment Shell
export> export COURSE=OSSP
Environment variable updated.
COURSE=OSSP
export>
```

Then test another variable:

`export COLLEGE=KLH`

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./export_variables
Export Environment Shell
export> export COURSE=OSSP
Environment variable updated.
COURSE=OSSP
export> export COLLEGE=KLH
Environment variable updated.
COLLEGE=KLH
export>
```

Step 19 — Finish the Export Program

Now type:

exit

Press Enter.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./export_variables
Export Environment Shell
export> export COURSE=OSSP
Environment variable updated.
COURSE=OSSP
export> export COLLEGE=KLH
Environment variable updated.
COLLEGE=KLH
export> exit
Export shell exited.
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

